

IMSciences, Peshawar

Database Systems Lab: Advanced Architecture Document

Geo-Spatial Disaster Relief & Supply Chain Optimizer

Project Team:

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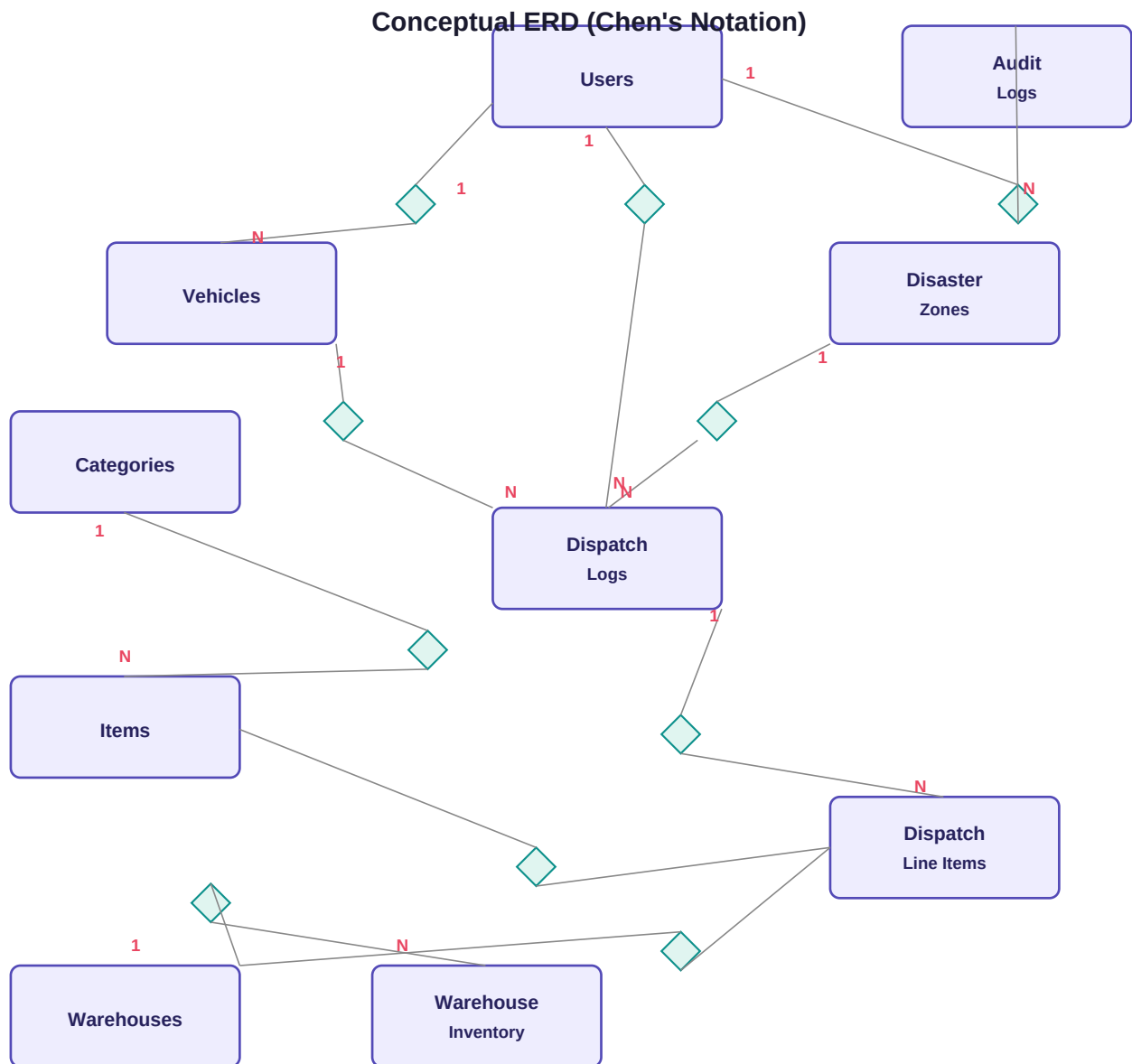
Instructor: Mr. Ali Hassan

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***Revision Notes:** Added Categories entity, corrected Audit_Logs schema with JSON change tracking, added ENUM constraint on Users.role, added business flow diagram, corrected all FK references.*

1. Conceptual ERD (Chen's Notation)

The diagram below models the normalized, production-grade schema for the Geo-Spatial Disaster Relief & Supply Chain Optimizer. The system tracks 10 entities across three functional domains: spatial and operational entities, inventory management, and dispatch and auditing. Primary Keys are underlined in the schema tables that follow. All relationships shown carry their cardinality constraints.



Note: The Audit_Logs entity connects to Users via changed_by (FK) and references all other tables generically through table_name and record_id, following the standard single-audit-table pattern. The Categories entity resolves the previously missing FK reference in Items.

2. Normalized Schema & Constraints

The database incorporates strict data validation, temporal tracking, and normalized M:N resolution. All ENUMs, CHECK constraints, UNIQUE constraints, and NOT NULL requirements are specified below. Primary keys are underlined; Foreign keys are marked with (FK).

2.1 Spatial & Operational Entities

Users	Data Type	Constraints
<u>user_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
username	VARCHAR(50)	UNIQUE, NOT NULL
email	VARCHAR(100)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	NOT NULL
role	ENUM	('Admin','Coordinator','Driver','Viewer'), NOT NULL
last_login	TIMESTAMP	DEFAULT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Warehouses	Data Type	Constraints
<u>warehouse_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	UNIQUE, NOT NULL
location	VARCHAR(200)	NOT NULL
latitude	DECIMAL(9,6)	NOT NULL
longitude	DECIMAL(9,6)	NOT NULL
capacity_kg	DECIMAL(10,2)	CHECK (capacity_kg > 0)
contact_person	VARCHAR(100)	DEFAULT NULL
is_active	BOOLEAN	DEFAULT TRUE

Disaster_Zones	Data Type	Constraints
<u>zone_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
description	TEXT	DEFAULT NULL
severity_level	ENUM	('Low','Medium','High','Critical'), NOT NULL
latitude	DECIMAL(9,6)	NOT NULL
longitude	DECIMAL(9,6)	NOT NULL
population_affected	INT	CHECK (population_affected >= 0)
is_active	BOOLEAN	DEFAULT TRUE
reported_by	INT	FK -> Users(user_id)
logged_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Vehicles	Data Type	Constraints
<u>vehicle_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
license_plate	VARCHAR(20)	UNIQUE, NOT NULL
type	VARCHAR(50)	NOT NULL
capacity_kg	DECIMAL(10,2)	CHECK (capacity_kg > 0)
status	ENUM	('Available','InTransit','Maintenance'), DEFAULT 'Available'
assigned_driver	INT	FK -> Users(user_id), DEFAULT NULL

2.2 Inventory Management (Normalized)

Categories	Data Type	Constraints
<u>category_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	UNIQUE, NOT NULL
description	TEXT	DEFAULT NULL

Items	Data Type	Constraints
<u>item_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	UNIQUE, NOT NULL
unit_of_measure	VARCHAR(20)	NOT NULL
weight_per_unit_kg	DECIMAL(8,3)	CHECK (weight_per_unit_kg > 0)
category_id	INT	FK -> Categories(category_id), NOT NULL

Warehouse_Inventory	Data Type	Constraints
<u>warehouse_id</u>	INT	PK (composite), FK -> Warehouses(warehouse_id)
<u>item_id</u>	INT	PK (composite), FK -> Items(item_id)
quantity_in_stock	INT	CHECK (quantity_in_stock >= 0), NOT NULL
last_updated	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE

Composite Primary Key (warehouse_id, item_id) resolves the M:N relationship between Warehouses and Items.

2.3 Dispatch & Auditing

Dispatch_Logs	Data Type	Constraints
<u>dispatch_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
zone_id	INT	FK -> Disaster_Zones(zone_id), NOT NULL
vehicle_id	INT	FK -> Vehicles(vehicle_id), NOT NULL
user_id	INT	FK -> Users(user_id), NOT NULL
priority	ENUM	('Low','Medium','High','Critical'), DEFAULT 'Medium'
status	ENUM	('Pending','EnRoute','Delivered','Failed'), DEFAULT 'Pending'
departure_time	TIMESTAMP	DEFAULT NULL
arrival_time	TIMESTAMP	DEFAULT NULL, CHECK (arrival_time > departure_time)
notes	TEXT	DEFAULT NULL

Dispatch_Line_Items	Data Type	Constraints
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<u>line_item_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
dispatch_id	INT	FK -> Dispatch_Logs(dispatch_id), NOT NULL
warehouse_id	INT	FK -> Warehouses(warehouse_id), NOT NULL
item_id	INT	FK -> Items(item_id), NOT NULL
quantity_dispatched	INT	CHECK (quantity_dispatched > 0), NOT NULL

Audit_Logs	Data Type	Constraints
<u>audit_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT
table_name	VARCHAR(50)	NOT NULL
record_id	INT	NOT NULL
action	ENUM	('INSERT','UPDATE','DELETE'), NOT NULL
changed_by	INT	FK -> Users(user_id)
changed_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
old_values	JSON	DEFAULT NULL (NULL for INSERT)
new_values	JSON	DEFAULT NULL (NULL for DELETE)

Audit_Logs uses a generic reference pattern: table_name identifies the source table, record_id points to the affected row's PK. The old_values and new_values JSON columns capture the complete state before and after each change, enabling full rollback capability.

3. Integrity Rules & Performance Strategies

3.1 Transactional Integrity (Preventing Over-Dispatch)

Strict database triggers are implemented on the `Dispatch_Line_Items` table. Before an `INSERT` is committed, a `BEFORE INSERT` trigger queries the `Warehouse_Inventory` table. If `quantity_dispatched` exceeds `quantity_in_stock`, the transaction is rolled back with a `SIGNAL SQLSTATE` raising an `InsufficientStock` error, ensuring absolute inventory accuracy. Additionally, a `CHECK` constraint on `Dispatch_Logs` ensures `arrival_time` cannot precede `departure_time`.

3.2 Performance & Indexing

Spatial Indexing: Standard B-Tree indexes are insufficient for geospatial proximity calculations. PostGIS GIST (Generalized Search Tree) indexes will be applied to the latitude and longitude columns of both `Warehouses` and `Disaster_Zones` to rapidly calculate the shortest route between a warehouse and a disaster zone.

Foreign Key Indexing: All foreign key columns are explicitly indexed to prevent full table scans during complex `JOIN` operations. This includes `zone_id`, `vehicle_id`, and `user_id` in `Dispatch_Logs`; `dispatch_id`, `warehouse_id`, and `item_id` in `Dispatch_Line_Items`; `category_id` in `Items`; and `reported_by` in `Disaster_Zones`.

3.3 Audit Trail & Priority Handling

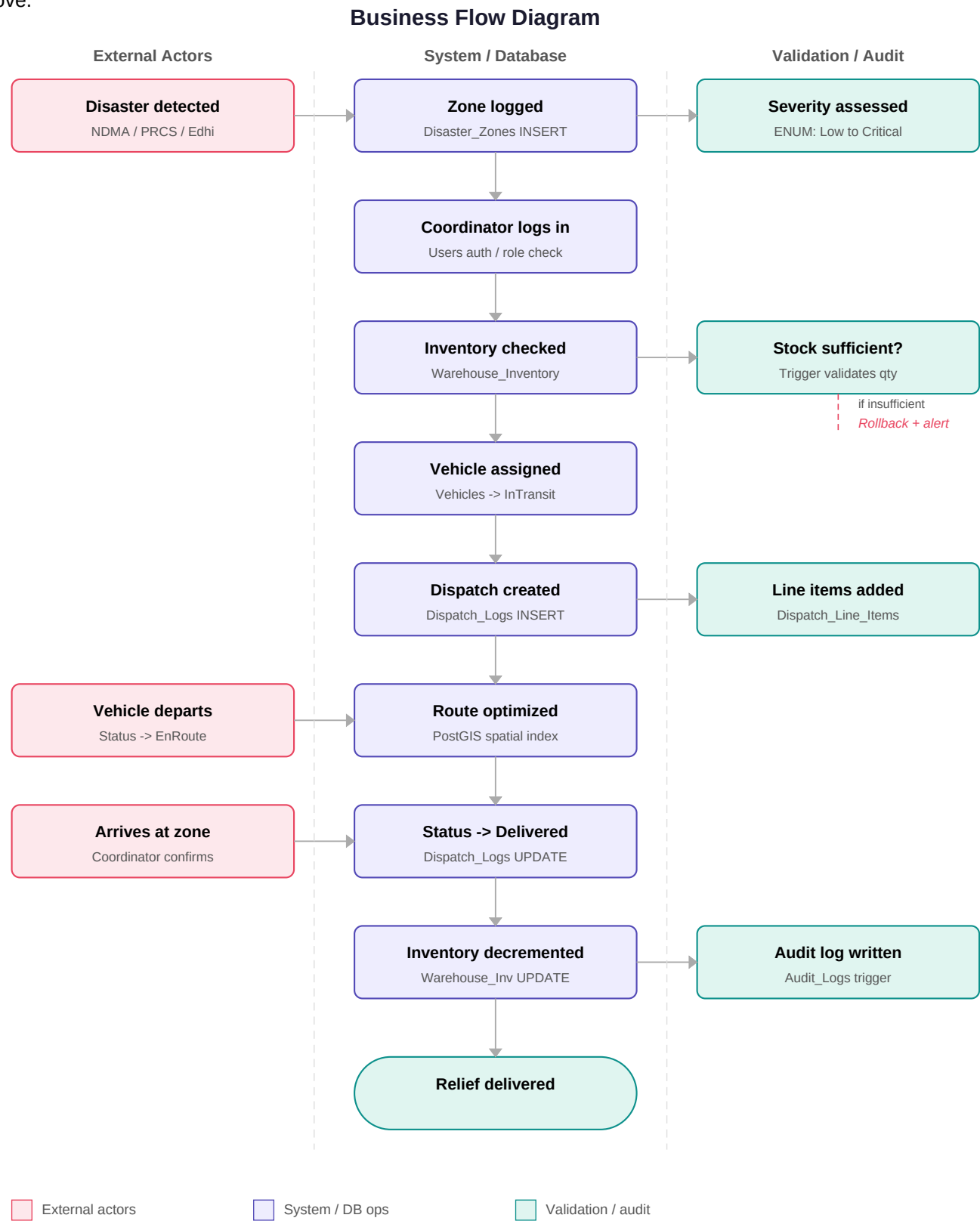
The `Audit_Logs` table serves as a permanent, append-only ledger for system accountability. Any `UPDATE` or `DELETE` on critical tables fires an automated trigger that captures the complete previous state as JSON in `old_values` and the new state in `new_values`, along with the responsible user. This enables full before/after comparison and forensic rollback capability. The `priority` attribute in `Dispatch_Logs` allows the system backend to route critical medical supplies before standard food rations based on disaster severity.

3.4 Role-Based Access Control

The `Users.role` column is constrained as an `ENUM` with four permitted values: `Admin`, `Coordinator`, `Driver`, and `Viewer`. Admin users manage warehouses and system configuration. Coordinators create and authorize dispatches. Drivers are assigned to vehicles and update delivery status. Viewers have read-only access for reporting and oversight. This enforces separation of duties at the schema level.

4. Business Flow Diagram

The diagram below illustrates the end-to-end operational flow of the disaster relief dispatch system. The process is organized into three swim lanes: External Actors (stakeholder organizations and field personnel), System/Database (core CRUD operations on the schema), and Validation/Audit (trigger-based integrity checks and audit logging). Each box maps directly to a table operation defined in the schema above.



4.1 Stakeholders

NDMA (National Disaster Management Authority): Government-level authority responsible for national disaster declaration and inter-agency coordination. Acts as the primary data source for disaster zone registration.

Pakistan Red Crescent Society (PRCS): Operates under the Geneva Conventions mandate with formal international federation backing (IFRC). Handles formal inter-agency dispatch coordination and resource allocation across provincial boundaries.

Edhi Foundation: Civilian rapid-response organization with extensive ground-level reach. Operates independently of government bureaucracy, enabling faster field deployment and last-mile delivery in remote disaster zones.

Warehouse & Logistics Coordinators: Operational users who manage inventory levels, authorize dispatches, and confirm deliveries. These are the primary system users interacting with the database on a daily basis.

5. Entity Summary

Entity	Domain	PK	FK References
Users	Operational	user_id	None
Warehouses	Spatial	warehouse_id	None
Disaster_Zones	Spatial	zone_id	Users (reported_by)
Vehicles	Operational	vehicle_id	Users (assigned_driver)
Categories	Inventory	category_id	None
Items	Inventory	item_id	Categories (category_id)
Warehouse_Inventory	Inventory	(warehouse_id, item_id)	Warehouses, Items
Dispatch_Logs	Dispatch	dispatch_id	Disaster_Zones, Vehicles, Users
Dispatch_Line_Items	Dispatch	line_item_id	Dispatch_Logs, Warehouses, Items
Audit_Logs	Auditing	audit_id	Users (changed_by)

Total: **10 entities**, **12 relationships**, **4 ENUM constraints**, **5 CHECK constraints**, **1 composite primary key**, **2 trigger-based validations** (stock check and audit logging).